The reduction of OEIS A355409 modulo k: a proof that the period divides $\varphi(k)$

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Abstract

OEIS A355409 carries a conjecture of P. Bala that for every positive integer k the sequence reduced modulo k is eventually periodic with period dividing $\varphi(k)$. It is true. The entry's own exponential generating function becomes, under the substitution $t = e^x - 1$, an ordinary power series with *integer* coefficients; that integrality forces all but finitely many terms of the Stirling expansion of $a(n)$ to vanish modulo k, and what is left is an integer combination of the functions $n \mapsto i^n$ with $0 \le i < k$. Each of those is eventually periodic with period dividing $\varphi(k)$, and so is the sum. The argument also gives the exact point at which periodicity starts: $n \ge v(k)$, where $v(k)$ is the largest exponent in the prime factorisation of k.

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1 The sequence and the conjecture

OEIS A355409 is "Expansion of e.g.f. 1/(1 + exp(2*x) - exp(3*x)).". It has offset 0 and begins

$$1, 1, 7, 55, 571, 7471, 117307, 2148175, \ \ldots$$

The entry states, not as a conjecture,

> Expansion of e.g.f. 1/(1 + exp(2*x) - exp(3*x)).

that is, writing $A(x) = \sum_{n\ge0} a(n)x^n/n!$,

$$A(x) \ = \ \frac{1}{1 + e^{2x} - e^{3x}}. \tag{1}$$

This is the only input taken from the entry, and it was checked against every one of the 19 terms of the entry's DATA section before use, not against a sample.

The entry separately carries the following comment:

> **Conjecture**: Let k be a positive integer. The sequence obtained by reducing a(n) modulo k is eventually periodic with period dividing phi(k) = A000010(k). For example, modulo 9 we obtain the sequence [1, 1, 7, 1, 4, 1, 1, 1, 7, 1, 4, 1, 1, 1, 7, 1, 4, 1, 1, ...] with an apparent period of 6 = phi(9) beginning at a(1). Cf. A354242. — *Peter Bala*, Apr 16 2024

As of the "Last modified" line on the live entry (Apr 19 2024 04:35:29, revision 19) the statement is still recorded as a conjecture, and nothing on the entry records it as settled either way.

Written out, the conjecture asserts: for every integer $k \ge 1$ there are n_0 and a positive integer P with $P \mid \varphi(k)$ such that

$$a(n + P) \ \equiv \ a(n) \pmod{k} \qquad \text{for all } n \ge n_0. \tag{2}$$

It is proved below, with $P = \varphi(k)$ and $n_0 = v(k)$, the largest exponent occurring in the prime factorisation of k.

2 The generating function in the variable $t = e^x - 1$

Set $t = e^x - 1$. Then $e^x = 1 + t$, and substituting into (1),

$$1 + e^{2x} - e^{3x} = 1 + (1+t)^2 - (1+t)^3 = 1 - t - 2t^2 - t^3,$$

so that

$$A(x) \;=\; G\big(e^x - 1\big), \qquad G(t) \;=\; \frac{1}{1 - t - 2t^2 - t^3}. \tag{3}$$

Lemma 1. *G is a power series in t with integer coefficients.*

Proof. G is a rational function with integer coefficients whose denominator has constant term 1. $\square$

Write $G(t) = \sum_{j \geq 0} c_j t^j$, so $c_j \in \mathbb{Z}$; here $c_j = 1, 1, 3, 6, 13, 28, 60, \ldots$, and the first few are

$$c_0, \ldots, c_8 \;=\; 1, 1, 3, 6, 13, 28, 60, 129, 277.$$

3 The Stirling expansion, and what survives modulo k

Let $S(n, j)$ denote the Stirling numbers of the second kind.

Lemma 2. *$a(n) = \sum_{j=0}^{n} c_j \, j! \, S(n, j)$ for every $n \geq 0$. In particular $a(n) \in \mathbb{Z}$.*

Proof. The standard expansion $\big(e^x - 1\big)^j = \sum_{n \geq j} j! \, S(n, j) \, x^n / n!$ holds because $j! \, S(n, j)$ counts the surjections from an n-set onto a j-set. Substituting it into $A(x) = G(e^x - 1) = \sum_j c_j \big(e^x - 1\big)^j$ and reading off the coefficient of $x^n/n!$ gives the claim; the sum is finite because $S(n, j) = 0$ for $j > n$. Integrality is then immediate from Lemma 1. $\square$

Lemma 3. *For every $k \geq 1$ and every n, $a(n) \equiv \sum_{j=0}^{k-1} c_j \, j! \, S(n, j) \pmod{k}$.*

Proof. For $j \geq k$ the integer $j!$ is a multiple of k, since k is one of the factors $1, 2, \ldots, j$. Hence every term of Lemma 2 with $j \geq k$ vanishes modulo k; this uses $c_j \in \mathbb{Z}$, which is Lemma 1. $\square$

This is the step at which integrality is not a convenience but the whole point: were the c_j merely rational, $c_j \, j!$ need not be divisible by k for large j and the sum would not truncate.

Lemma 4. *There are integers $d_0, \ldots, d_{k-1}$, namely $d_i = \sum_{j=i}^{k-1} (-1)^{j-i} \binom{j}{i} c_j$, with*

$$a(n) \;\equiv\; \sum_{i=0}^{k-1} d_i \, i^n \pmod{k} \qquad \text{for all } n \geq 0.$$

Proof. Counting surjections from an n-set onto a j-set by inclusion and exclusion on the image gives $j! \, S(n, j) = \sum_{i=0}^{j} (-1)^{j-i} \binom{j}{i} i^n$. Substituting this into Lemma 3 and exchanging the two finite sums collects the coefficient of i^n into d_i as stated. Each d_i is an integer because each c_j is. $\square$

4 The period

For $k \geq 2$ let $v(k)$ be the largest exponent in the prime factorisation of k, so $k \mid m^{v(k)}$ whenever m is divisible by every prime factor of k, and $v(k) \leq \log_2 k$.

Lemma 5. *Let $k \geq 2$ and $0 \leq i < k$. Then $i^{\,n+\varphi(k)} \equiv i^{\,n} \pmod{k}$ for every $n \geq v(k)$.*

Proof. Write $k = k_1 k_2$, where $k_1 = \prod_{p \mid k,\ p \mid i} p^{v_p(k)}$ and $k_2 = k/k_1$; then $\gcd(k_1, k_2) = 1$ and $\gcd(i, k_2) = 1$.

Modulo k_1: for each prime $p \mid k_1$ we have $p \mid i$, so $v_p\big(i^n\big) = n\, v_p(i) \geq n \geq v(k) \geq v_p(k)$. Hence $k_1 \mid i^n$ for $n \geq v(k)$, and likewise $k_1 \mid i^{n+\varphi(k)}$; both sides are $\equiv 0$.

Modulo k_2: i is a unit, so by Euler's theorem $i^{\varphi(k_2)} \equiv 1$. Since $\gcd(k_1, k_2) = 1$ we have $\varphi(k) = \varphi(k_1)\varphi(k_2)$, so $\varphi(k_2) \mid \varphi(k)$ and therefore $i^{\varphi(k)} \equiv 1 \pmod{k_2}$, giving $i^{\,n+\varphi(k)} \equiv i^{\,n}$.

The two congruences and the Chinese remainder theorem give the claim modulo $k = k_1 k_2$. (For $i = 0$ note $0^n = 0$ for $n \geq 1$ and $v(k) \geq 1$.) ☐

Theorem 6. *Let $a(n)$ be OEIS A355409. For every integer $k \geq 1$,*

$$a\big(n + \varphi(k)\big) \;\equiv\; a(n) \pmod{k} \qquad \textit{for all } n \geq v(k),$$

where $v(k)$ is the largest exponent in the prime factorisation of k (and $v(1) = 0$). In particular the sequence $\big(a(n) \bmod k\big)_{n\geq 0}$ is eventually periodic with period dividing $\varphi(k)$, with a pre-period of length at most $\log_2 k$. This is the conjecture of Section 1.

Proof. The case $k = 1$ is trivial. For $k \geq 2$, Lemma 4 writes $a(n)$ modulo k as $\sum_{i<k} d_i i^n$ with integer d_i independent of n, and Lemma 5 gives $i^{\,n+\varphi(k)} \equiv i^{\,n}$ for each $i < k$ once $n \geq v(k)$. Summing the k congruences with the weights d_i gives $a(n+\varphi(k)) \equiv a(n) \pmod{k}$ for $n \geq v(k)$. The true period of the eventually periodic sequence is then a divisor of $\varphi(k)$. ☐

Remark 7. The bound $n \geq v(k)$ cannot be lowered in general: for k a prime power p^e the term $i = p$ contributes p^n, which is not divisible by p^e until $n = e = v(k)$.

5 Verification

Three checks, each independent of the algebra above.

First, the series G of (3) was expanded and $\sum_j c_j j! S(n,j)$ compared against the entry's DATA at every one of the 19 published indices; the values agree exactly, and every c_j came out an integer. A series that reproduced only some of the published terms would not have been used.

Second, for each k from 2 to 40 the sequence $a(n) \bmod k$ was computed exactly — not sampled — from the recurrence $S(n,j) = j\, S(n-1,j) + S(n-1,j-1)$ carried out in $\mathbb{Z}/k\mathbb{Z}$ on the truncated state $\big(S(n,0), \ldots, S(n,k-1)\big)$ of Lemma 3, and its exact eventual period found by detecting the repetition of that state. The result is tabulated below. In every case the period divides $\varphi(k)$, as Theorem 6 requires.

k	$\varphi(k)$	period	k	$\varphi(k)$	period
2	1	1	22	10	10
3	2	1	23	22	22
4	2	1	24	8	2
5	4	4	25	20	20
6	2	1	26	12	12
7	6	6	27	18	18
8	4	2	28	12	6
9	6	6	29	28	28
10	4	4	30	8	4
11	10	10	31	30	30
12	4	1	32	16	4
13	12	12	33	20	10
14	6	6	34	16	16
15	8	4	35	24	12
16	8	2	36	12	6
17	16	16	37	36	36
18	6	6	38	18	18
19	18	18	39	24	12
20	8	4	40	16	4
21	12	6			

Third, the congruence $a(n+\varphi(k)) \equiv a(n) \pmod{k}$ of Theorem 6 was checked directly on the published terms for every k in that range and every $n \geq v(k)$ for which both indices are published; it holds without exception.

References

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